Foundational Mathematics & Mechanics of Digital Images

1. Mathematical Definition of an Image

- **The 2D Representation:** Mathematically, an image is described as a two-dimensional function *f(x, y)*.
- *(x, y)*: Represents the spatial coordinates (the location of a pixel in a 2D grid).
- *f* (Intensity): The value of the function *f* at any given point *(x, y)* is directly proportional to the brightness or gray level of the image at that specific point.
- **Spatial Bounds:** For an image with a height *h* and a width *w*:
  - $x \in$ [0, h-1] (Vertical bounds / rows)
  - $y \in$ [0, w-1] (Horizontal bounds / columns)
- **Discrete Intensity Bound:** For a standard 8-bit image, the pixel intensity function is bounded by $f(x, y) \in [0, L-1]$, where L = 256.

2. The Mechanics of Intensity (Why 256 Values?)

- **Binary Unit Core:** Data inside a computer is binary.
  - 1 bit can hold 2^1 variations (either 0 or 1).
  - N bits can hold 2^N variations.
- **The 8-Bit Standard:** Digital images traditionally allocate 8 bits (1 byte) to represent a single pixel or a single color channel.
- **The Calculation:** Total Values = 2^8 = 256 discrete intensity steps (ranging strictly from 0 to 255).

3. Analog vs. Digital Image

- **Analog Image:** In the real physical world, light capture is a continuous function. Both spatial points *(x, y)* and brightness values are infinite and unbroken.
- **Digital Image:** A digital representation converts the real world into discrete values via spatial sampling (creating individual pixels) and quantization (mapping brightness levels strictly to integer steps like 0–255).

4. The Additive RGB Color Model

- **Additive Principle:** Color images form a broad gamut by superimposing light intensities of three primary colors: Red (R), Green (G), and Blue (B).
- **Primary Intersections:**
  - Red + Green = Yellow

    - Red + Blue = Magenta
    - Green + Blue = Cyan
    - Red + Green + Blue = White
- **Channel Implementations:** A single pixel color requires three parallel functional values representing each color channel: $F_R(x,y) \in [0, 255]$, $F_G(x,y) \in [0, 255]$, $F_B(x,y) \in [0, 255]$.
- **Color Depth Definition:** The total number of bits utilized to indicate the final color variety of a single image pixel.
    - Since each individual channel (R, G, B) requires 8 bits: Total Color Depth = 3 channels × 8 bits = 24 bits per pixel.

5. Grayscale vs. Black & White (Binary) Images

- **Grayscale Images:** Utilize 1 functional channel mapping brightness intensity. It contains the complete spectrum of shades from deep black to bright white across 256 distinct gray values (0 to 255).
- **Black & White (Binary) Images:** Highly restrictive matrix containing only 2 extreme values. A pixel intensity is either completely black (0) or completely white (255).

6. Storage Memory Calculations (Data Math)

Formula for Raw Uncompressed Image Size:

Size (in Bytes) = Width × Height × Number of Channels

Example: Calculating uncompressed memory for a 1000 × 600 pixel layout:

Image Type	Channels Required	Bit Depth per Pixel	Total Raw Size Calculation	Total Bytes
Grayscale	1 Channel	8 bits (1 Byte)	1000 × 600 × 1 Byte	600,000 Bytes (~600 KB)
RGB Color	3 Channels	24 bits (3 Bytes)	1000 × 600 × 3 Bytes	1,800,000 Bytes (~1.8 MB)

Note: In practice, file extensions like JPEG or PNG deploy coding compression schemes to shrink these numbers down significantly before saving to disk.